

MASENO 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

232/2

PHYSICS

PAPER 2 (THEORY)

TIME: 2 HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

INSTRUCTIONS TO CANDIDATES

- (a) Write your name, Index number and School in the spaces provided above
- (b) Sign and write the date of the examination in the spaces provided above
- (c) This paper consists of two sections; A and B
- (d) Answer all the questions in section A and B in the spaces provided
- (e) All workings must be clearly shown in the spaces provided below each question
- (f) Non programmable silent electronic calculators may be used.
- (g) Candidates should answer the questions in English

FOR EXAMINER'S USE ONLY

Section	Question	Maximum Score	Candidate's Score
A	1 - 13	25	
B	14	11	
	15	10	
	16	12	
	17	10	
	18	12	
Total Score		80	

SECTION A (25marks)

Answer **all** the questions in this section in the spaces provided

1. **Figure 1** below shows the set – up for a simple cell.

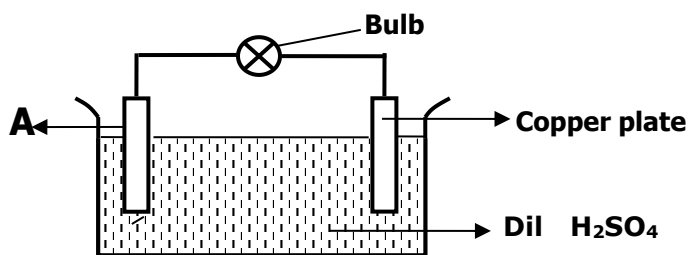


Figure 1.

(i) Explain how electrode **A** becomes **negative**
(1mk)

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(ii) Explain why the bulb goes off after only a short time. (1mk)

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2. a) **Figure 2** shows an object, a screen and light sources X, Y and Z.

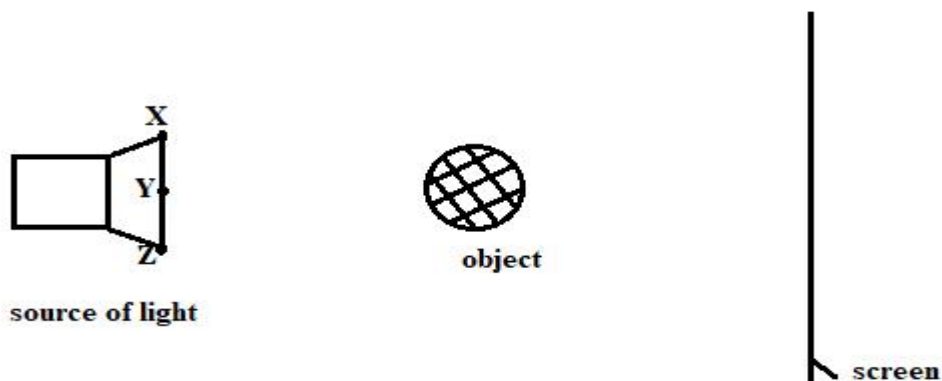


Figure 2.

i) Complete the diagram to show the formation of a shadow (2mks)

ii) State one property of the object that makes it possible for its shadow to be formed (1mk)

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3. A highly negatively charged rod is gradually brought close to the cap of a positively charged electroscope. It is observed that the leaf collapses initially and then diverges. Explain the observation.

(2mks)

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4. Distinguish between thermionic emission and photoelectric effect. (1mk)

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5. Explain why mains electricity is transmitted through alternating current and not direct Current

(1mk)

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6. **Figure 3** shows two mirrors inclined at an angle of 30° to each other. A ray of light is incident on mirror M_1 as shown

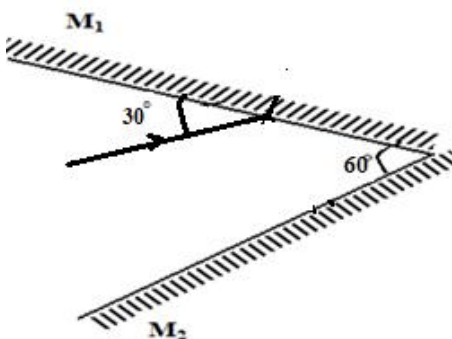


Figure 3.

Sketch the path of the ray to show its reflection (1mk)

7. **Figure 4** below shows sound waves in air produced by a vibrating tuning fork. R is an air molecule on the path of the waves.

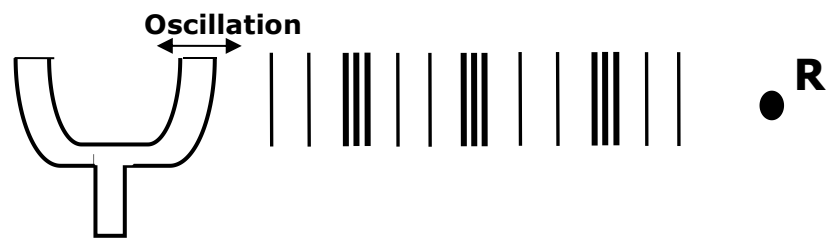


Figure 4.

- i) Using a line, indicate on the diagram a distance **d** equal to one wavelength of the wave. **(1mk)**

- ii) In the space provided below, show with an arrow the direction of motion of the air molecule **R** as the waves pass. **(1mk)**

8. **Figure 5** below shows a bar magnet attracting steel pin as shown

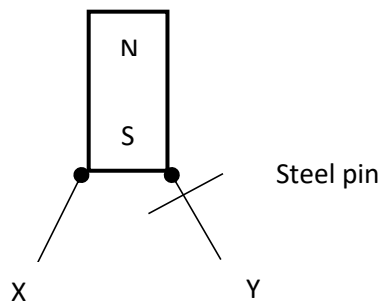


Figure 5

State and explain what would happen when a North pole of a bar magnet is brought near the tips of steel pin X and Y. **(2mks)**

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9. Figure 6 below shows a wave profile for a wave whose frequency is 5Hz.

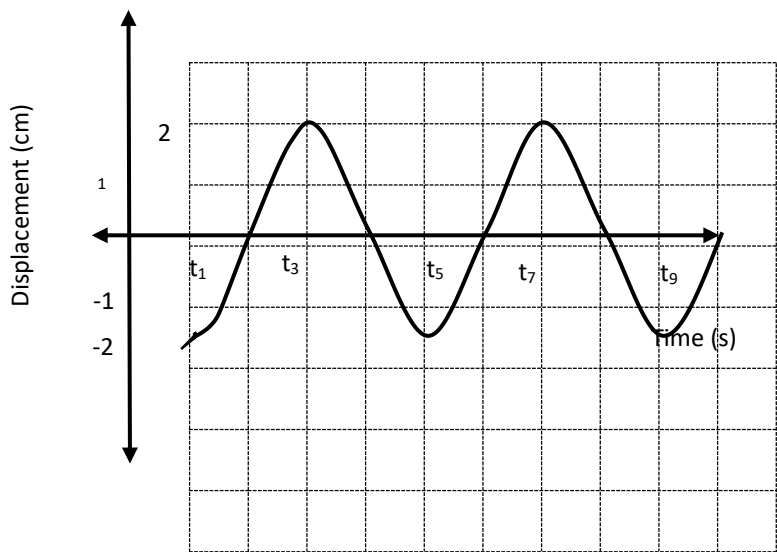


Figure 6

Determine the value of t_8 . (2mks)

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10. Figure 7 below shows circular waves originating from the principal focus F of a concave mirror and moving towards the mirror.

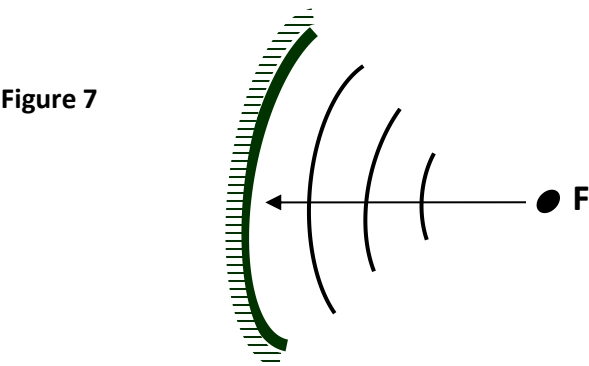


Figure 7

Complete the diagram to show the reflected waves. (1mk)

11. Two heating coils A and B connected in parallel in a circuit produces power of 36W and 54W respectively. What is the ratio of the resistance of B to that of A (3mks)

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12. Figure 8 below shows the path of light through a transparent material placed in air.

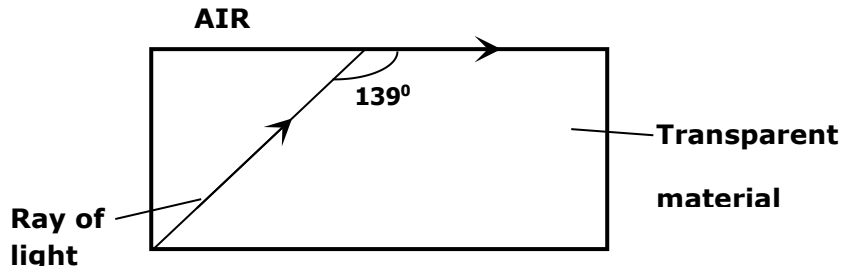


Figure 8.

Calculate the refractive index of the transparent material. (3mks)

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13. Figure 9 below shows an image; I formed by an object placed in front of a *convex mirror*. On the diagram draw appropriate rays and locate the position of the object. (2mks)

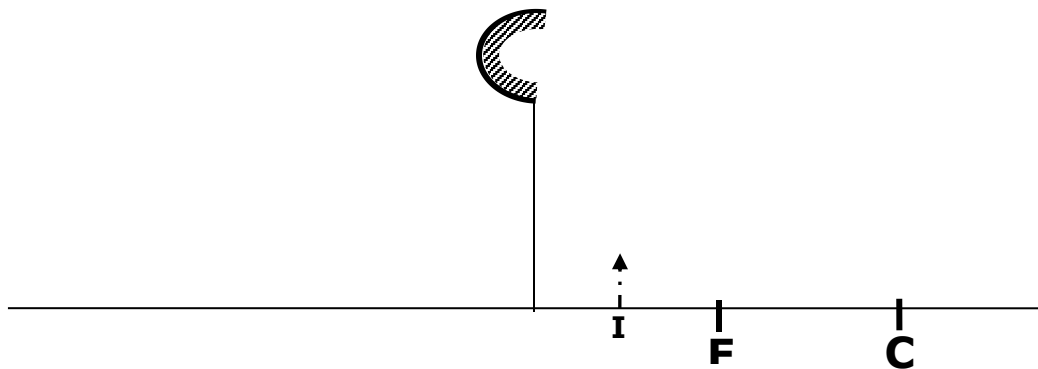


Figure 9.

SECTION B (55 marks)

Answer all the questions in this section in the spaces provided

14(a) Figure 10 below shows two charged plates close to each other

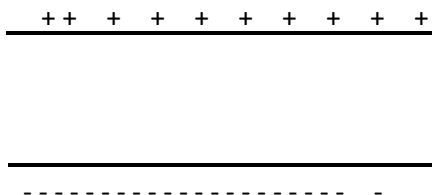


Figure 10.

(i) Complete the diagram to show the electric field patterns between the plates **(1 mark)**

(ii) Without changing the area of overlap, suggest any one way of increasing the charge stored per unit voltage on the plates **(2 mks)**

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(b) Figure 11 shows two coils P and Q placed close to each other. When the switch **S** is closed, an e.m.f is induced in coil Q. Similarly, an e.m.f is induced in coil Q when the switch **S** is opened.

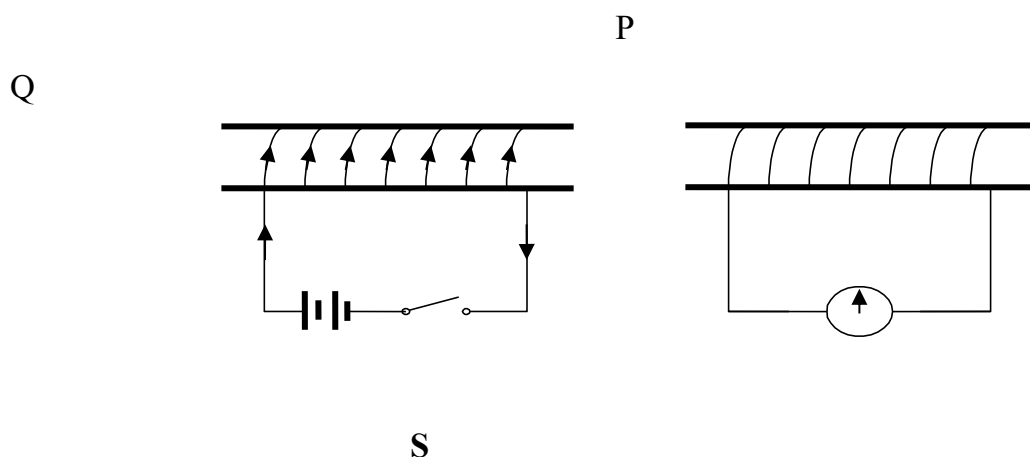


Figure 11.

Explain why the induced current in coil Q is higher when the switch **S** in coil P is opened than when it is closed. **(2 mks)**

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(b) The diagram in **Figure 12** below shows an induction coil used to produce sparks.

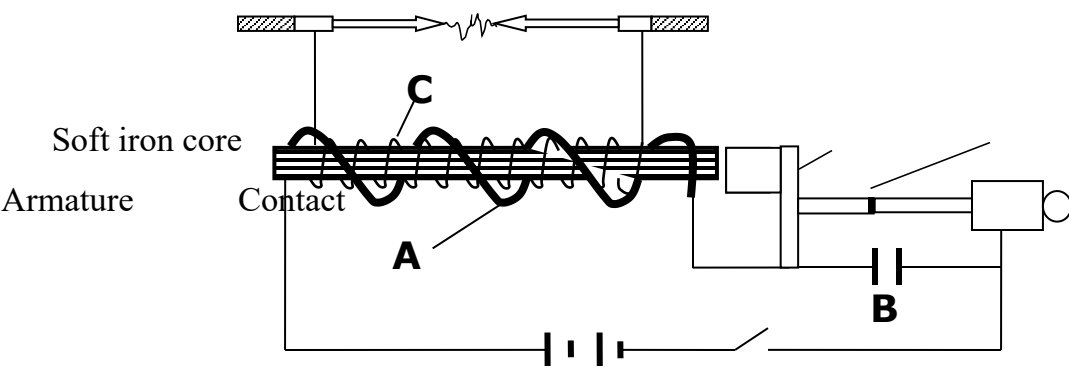


Figure 12.

(i) Name parts labeled A, and C (2mks)

A.....

C.....

(ii) Briefly explain how the induction coil works. (3mks)

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(c) State the function of part B in the diagram. (1mk)

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15. (a) **Figure 13** below shows a graph of resistance against reciprocal of current. Use it to answer the questions that follow.

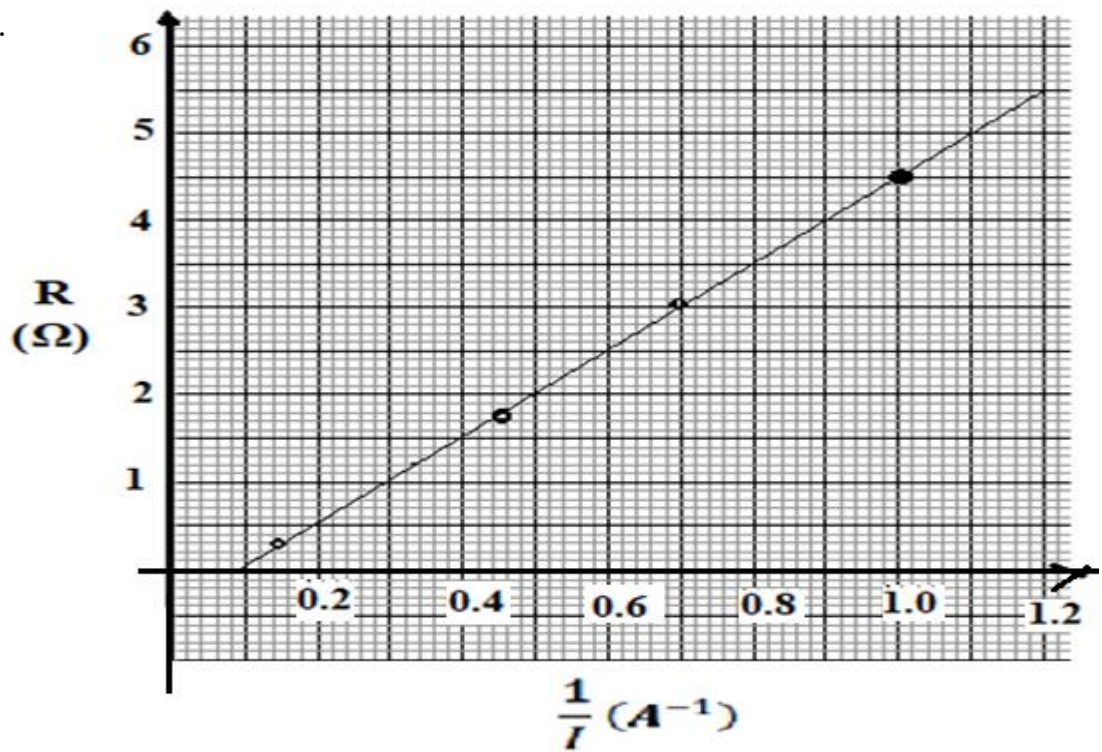


Figure 13

i) Find the internal resistance, r . (2mks)

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ii) Determine the e.m.f, E of the cell. (2mks)

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(b) Three resistors of resistance $2\ \Omega$, $3\ \Omega$ and $4\ \Omega$, are to be connected to a cell such that they have the least effective resistance.

(i) Draw a circuit to show how they can be connected to achieve this. (1mk)

(ii) Determine the least effective resistance of the three resistors. (2mks)

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(c) An electric iron rated 240V, 750w is to be connected to a 240v mains supply through a 3A fuse.
Determine whether the fuse is suitable or not. (3mks)

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16. (a) (i) Name the metal used to shield X-rays operators from the radiation. Give reasons why it is used. (2mks)

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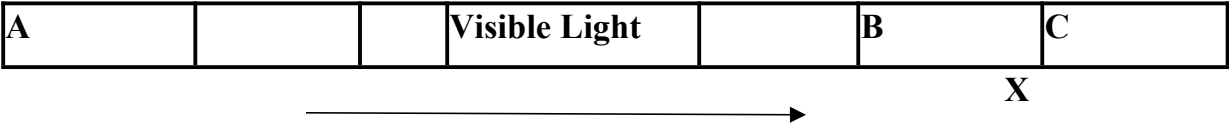
(ii)In a television set, magnetic fields are preferred for use as deflection system instead of electric fields. Explain. (2mks)

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(b) Figure 14 below show regions of the complete electromagnetic spectrum.



Y

Figure 14.

Decreasing frequency

(i) Name the colour of light at X and Y

(2mks)

X

Y

(ii) Name a suitable detector for radiation B

(1mk)

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(c) A student connected a circuit as shown in **Figure 15** below, hoping to produce a rectified output.

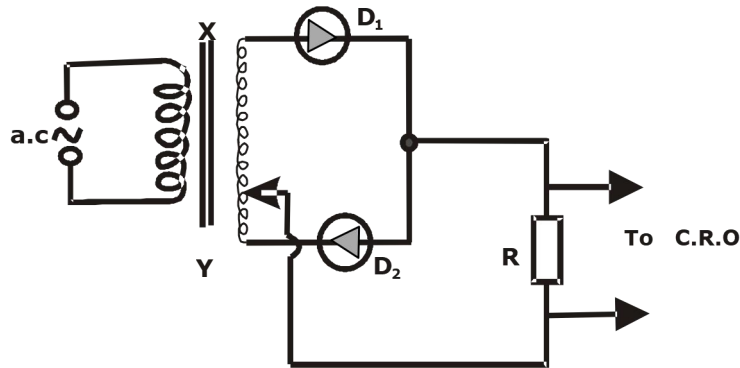


Figure 15.

(i) Sketch the graph of the output on the CRO screen in the space below.

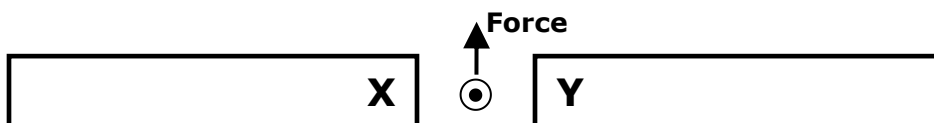
(1mk)

(ii) Explain how this output is produced

(2mks)

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(d) The figure shows a current – carrying conductor in a magnetic field direction of force on the wire is as shown by the arrow.

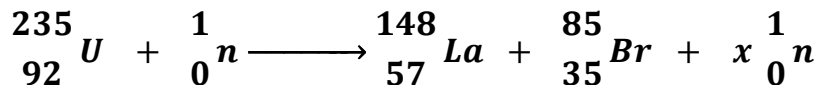


State the polarities of A and B. (2mks)

X

Y.

17. (a) A typical nuclear fission reaction in a nuclear reactor is given below.



(i) What is meant by nuclear fusion? (1mk)

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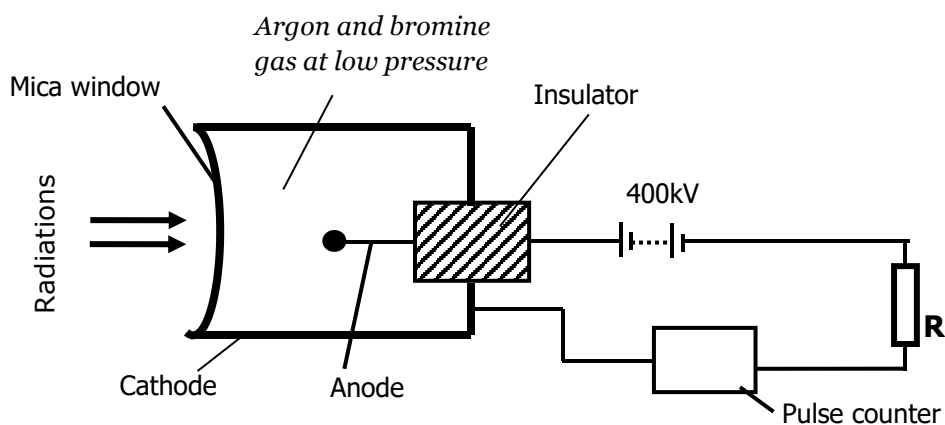
(ii) Find the value of x. (1mk)

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(iii) How are the neutrons produced used in the reactor? (1mk)

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(b) Figure 16 below shows the diagram of a Geiger – Muller tube connected to a power supply and a pulse counter.



(i) Why should the Argon gas be at low pressure? (1mk)

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(ii) State the purpose of the bromine gas in the tube (1mk)

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(iii) Suggest one way of increasing the sensitivity of the tube (1mk)

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(c) Figure 17 below shows an arrangement used to investigate photoelectric effect

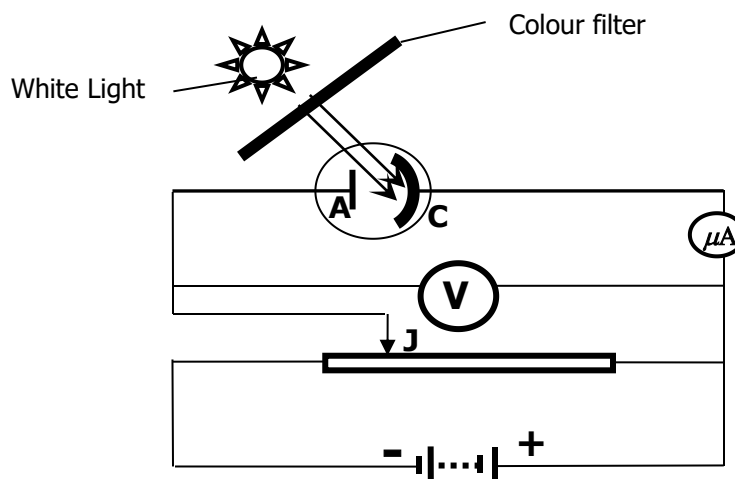


Figure 17

i) What is the purpose of the colour filter? (1mk)

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ii) State two measurable quantities in this set up (2mks)

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iii) State how the intensity of light affects photo current (1mk)

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18. (a) Under what conditions does a converging lens form Virtual images (1 mk)

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(b) Sketch on a diagram to illustrate how a convex lens is used as a magnifying glass.3mks)

(c)The diagram in **Figure 18** below shows a three-pin plug.

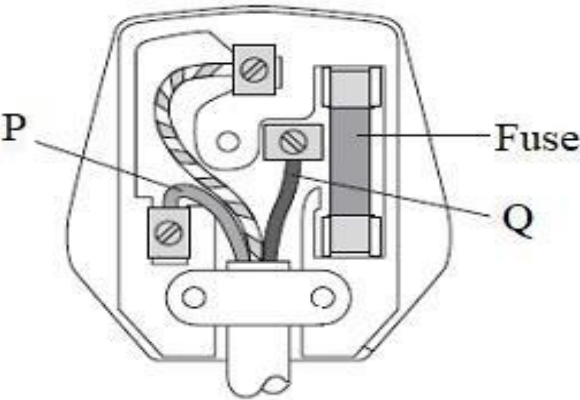


Figure 18.

(i) Name the colour of conductors P and Q (2 marks)

P

Q

(ii) Why is the earth pin longer than the rest in the three-pin plug shown above? (1 mark)

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(b) Figure 19 shows the displacement of a practice in progressive wave incident on a boundary between deep and shallow regions.

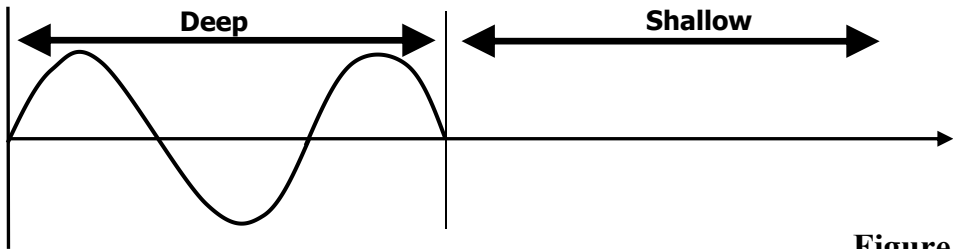


Figure 19.

(i) Complete the diagram to show what is observed after boundary. (1mk)

(ii) Explain the observation in (i) above. (1mk)

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(c) Concave mirrors are used by dentists to examine teeth. By use of a ray diagram show how this is achieved. (3 marks)